TECHNICAL REPORT: SPC & PROCESS CAPABILITY (Cp & Cpk)

Day-4 Quality Engineering Challenge — 5-Axis CNC Blade Grinding

Component Name: High Pressure Gas Turbine Blade
Material: Inconel 718 (Superalloy)
Parameter Checked: Blade Root Thickness
Lead Inspector: Jagriti Singh (Quality Engineer)
Target Industry: Aerospace Manufacturing
Target Specification: USL = 12.515 mm | LSL = 12.485 mm

1. Document Details & Objective

Target Industry & Application: Aerospace Industry (Precision Superalloy Component Machining).

Objectives of the Study:

- To check whether the 5-Axis CNC machine is manufacturing components to the correct dimension/size or not.
- To evaluate whether our precision grinding process is operating strictly within the customer-provided engineering specifications.
- To perform Statistical Process Control (SPC) charting (X-bar & R chart) and assess Process Capability (Cp, Cpk) against aerospace benchmark thresholds (Cpk ≥ 1.33).

2. Sample Data Collection Table

Subgroup	Sample 1 (mm)	Sample 2 (mm)	Sample 3 (mm)	Sample 4 (mm)	Mean (X-bar) (mm)	Range (R) (mm)
SG-1	12.502	12.505	12.498	12.501	**12.5015**	**0.007**
SG-2	12.499	12.503	12.502	12.500	**12.5010**	**0.004**
SG-3	12.506	12.501	12.497	12.504	**12.5020**	**0.009**
SG-4	12.496	12.500	12.501	12.499	**12.4995**	**0.003**
SG-5	12.503	12.502	12.504	12.501	**12.5025**	**0.003**

3. Process Averages & Standard Deviation

1. Process Grand Mean (X-double-bar):

```
X-double-bar = (12.5015 + 12.5010 + 12.5020 + 12.4995 + 12.5025) / 5 = 12.5013 mm
```

2. Average Range (R-bar):

```
R-bar = (0.007 + 0.004 + 0.009 + 0.003 + 0.003) / 5 = 0.0052 mm
```

3. Standard Constants (for subgroup size n = 4):

A2 = 0.729 | d2 = 2.059 | D3 = 0 | D4 = 2.282

4. Standard Deviation Estimation (σ):

```
σ = R-bar / d2 = 0.0052 / 2.059 = 0.00252 mm
```

4. Process Capability Calculations (Cp & Cpk)

1. Process Capability Index (Cp):

```
Cp = (USL - LSL) / (6 × σ) = (12.515 - 12.485) / (6 × 0.00252) = 0.030 / 0.01512 = 1.98
```

2. Upper Capability Index (Cpu):

```
Cpu = (USL - X-double-bar) / (3 × σ) = (12.515 - 12.5013) / (3 × 0.00252) = 0.0137 / 0.00756 =
1.81
```

3. Lower Capability Index (Cpl):

```
Cpl = (X-double-bar - LSL) / (3 × σ) = (12.5013 - 12.485) / (3 × 0.00252) = 0.0163 / 0.00756 = 2.15
```

4. Critical Capability Index (Cpk):

```
Cpk = min(Cpu, Cpl) = min(1.81, 2.15) = 1.81
```

Cp Index	Cpu Index	Cpl Index	Final Cpk
1.98	1.81	2.15	1.81

5. Control Limits for X-bar & R Charts

A) X-bar Chart Control Limits:

- Center Line (CL): 12.5013 mm
- Upper Control Limit (UCL): X-double-bar + (A2 × R-bar) = 12.5013 + 0.00379 = 12.5052 mm
- Lower Control Limit (LCL): X-double-bar - (A2 × R-bar) = 12.5013 - 0.00379 = 12.4945 mm

B) R Chart (Range Chart) Control Limits:

- Center Line (CL): 0.0052 mm
- Upper Control Limit (UCL): D4 × R-bar = 2.282 × 0.0052 = 0.0118 mm
- Lower Control Limit (LCL): D3 × R-bar = 0 × 0.0052 = 0 mm

6. Final Observations & Engineering Conclusions

1. **Process Capability Status:** Cpk = 1.81 (> 1.33 benchmark). The CNC grinding process is **highly capable** of meeting aerospace quality standards.
2. **Process Centering:** Process Mean (12.5013 mm) is virtually centered on the target dimension (12.500 mm).
3. **Statistical Control Status: PROCESS IN STATISTICAL CONTROL**. All subgroup means and ranges remain within control limits with no out-of-control signals.